Q7.6.) 중회귀모형에서 다음을 증명하라.

(1) $Cov(e, y) = [I - \Pi_X]\sigma^2$ (2) $Cov(e, \hat{y}) = 0$

(3) $Cov(e, b) = 0$ (4) $Cov(\epsilon, b) = X(X'X)^{-1}\sigma^2$

(5) $\sum_{j=1}^{n} e_j y_j = SSE$ (6) $\sum_{j=1}^{n} e_j \hat{y}_j = 0$

* $e = y - \hat{y}$, $b = \hat{\beta} = (X^TX)^{-1}X^Ty$

또한 절편 없는 중회귀모형에서는 어떻게 되는가를 연구하여라.

(1) $Cov(e, y) = Cov(y-\hat{y}, y) = Cov((I_n - \Pi_X)y, y) = (I_n - \Pi_X)Var(y)$
$= \sigma^2(I_n - \Pi_X)$

(2) $Cov(e, \hat{y}) = Cov(y - \hat{y}, \hat{y}) = Cov((I_n - \Pi_X)y, \Pi_X y)$
$= y^T(I_n - \Pi_X)\Pi_X y = 0$
$\because I_n - \Pi_X$ and Π_X are orthogonal

(3) $Cov(e, b) = Cov(y-\hat{y}, (X^TX)^{-1}X^Ty)$
$= Cov(y-\hat{y}, y)X(X^TX)^{-1} = 0 \quad \because (1)$

(4) $Cov(\epsilon, b) = Cov(\epsilon, (X^TX)^{-1}X^Ty) = Cov(\epsilon, y)X(X^TX)^{-1} = Cov(\epsilon, X\beta + \epsilon)X(X^TX)^{-1}$
$= Var(\epsilon)X(X^TX)^{-1} = \sigma^2 I_n X(X^TX)^{-1} = \sigma^2 X(X^TX)^{-1}$

(5) $\sum_{j=1}^{n} e_j y_j = \sum_{j=1}^{n}(y_j - \hat{y}_j)y_j = [(I_n - \Pi_X)y]^T y = y^T(I_n - \Pi_X)y = SSE$

(6) $\sum_{j=1}^{n} e_j \hat{y}_j = \sum_{j=1}^{n}(y_j - \hat{y}_j)\hat{y}_j = [(I_n - \Pi_X)y]^T(\Pi_X y) = y^T(I_n - \Pi_X)\Pi_X y = 0$
$\because (I_n - \Pi_X)\Pi_X = 0$

절편 없는 중회귀모형 : $y = X\beta$, $X = [x_1 \cdots x_p]$, $\beta = [\beta_1 \cdots \beta_p]^T$

Q7.7.) 오차의 제곱합이 다음과 같이 됨을 보이고 $\beta = b$ 에서 왼쪽의 오차제곱합이 최소가 됨을 말하라.

$$(y - X\beta)'(y - X\beta) = (y - Xb)'(y - Xb) + (\beta - b)'X'X(\beta - b)$$

$(y - X\beta)'(y - X\beta) = (y - Xb + Xb - X\beta)'(y - Xb + Xb - X\beta)$
$= (y - Xb)'(y - Xb) + (Xb - X\beta)'(Xb - X\beta) + 2(Xb - X\beta)'(y - Xb)$

$(Xb - X\beta)'(y - Xb) = (b - \beta)'X'(I_n - \Pi_X)y = 0 \quad \because X'(I_n - \Pi_X) = 0$

$\therefore (y - X\beta)'(y - X\beta) = (y - Xb)'(y - Xb) + (b - \beta)'X'X(b - \beta)$
$\geq (y - Xb)'(y - Xb)$
$(\because (b-\beta)'X'X(b-\beta) > 0$ since $X'X$ is positive definite$)$

$\therefore \quad b = \underset{\beta \in \mathbb{R}^{p+1}}{\operatorname{argmin}} (y - X\beta)'(y - X\beta)$

Q7.8.) 어떤 알려진 상수 a_j, b_j 에 대하여

$$y_j = a_j\beta_1 + b_j\beta_2 + \epsilon_j, \quad j = 1, 2, \cdots, n$$

이고 ϵ_j 는 서로 독립이며 $N(0, \sigma^2)$ 의 분포에 따른다고 하자. β_1 과 β_2 의 최소제곱추정량이 서로 독립일 필요충분조건을 찾아라.

$y = [y_1 \cdots y_n]^T$, $X = [X_1 \; X_2]$, $X_1 = [a_1 \cdots a_n]^T$, $X_2 = [b_1 \cdots b_n]^T$
$\beta = [\beta_1 \; \beta_2]^T$

$y = X\beta = X_1\beta_1 + X_2\beta_2$
$\hat{y} = \Pi_X y = \Pi_{X_1}y + \Pi_{X_{2,1}}y$, $\Pi_P = P(P'P)^{-1}P'$
$= X_1\hat{\beta}_1 + X_2\hat{\beta}_2 = X_1\hat{\beta}_1 + \Pi_{X_1}X_2\hat{\beta}_2 + X_{2,1}\hat{\beta}_2$, $X_{2,1} = X_2 - \Pi_{X_1}X_2$

$\therefore \Pi_{X_1}y = X_1\hat{\beta}_1 + \Pi_{X_1}X_2\hat{\beta}_2$ and $\Pi_{X_{2,1}}y = X_{2,1}\hat{\beta}_2$
$X_1\hat{\beta}_1 = \Pi_{X_1}y - \Pi_{X_1}X_2\hat{\beta}_2$ and $\hat{\beta}_2 = (X_{2,1}'X_{2,1})^{-1}X_{2,1}'y$

if $\hat{\beta}_1$ and $\hat{\beta}_2$ are independent $\Leftrightarrow X_1^TX_2 = 0$

[Textbook Approach]

$$X'X = \begin{bmatrix} \sum a_i^2 & \sum a_ib_i \\ \sum a_ib_i & \sum b_i^2 \end{bmatrix}$$

$$(X'X)^{-1}\sigma^2 = \frac{\sigma^2}{(\sum a_i^2)(\sum b_i^2) - (\sum a_ib_i)^2}\begin{bmatrix} \sum b_i^2 & -\sum a_ib_i \\ -\sum a_ib_i & \sum a_i^2 \end{bmatrix}$$

따라서 필요충분조건은 $\sum a_ib_i = 0$

Q7.9.) 만약

$y_1 = \beta_1 + \epsilon_1 \quad y_2 = 2\beta_1 - \beta_2 + \epsilon_2 \quad y_3 \sim \beta_1 + 2\beta_2 + \epsilon_3 \quad \epsilon_i \sim N(0, \sigma^2 I)$

$$y = X\beta + \epsilon \quad \beta = [\beta_1 \; \beta_2]^T, \; X = \begin{bmatrix} 1 & 0 \\ 2 & -1 \\ 1 & 2 \end{bmatrix} = [X_1 \; X_2]$$

이라고 하자. 가설 $H_0 : \beta_1 = \beta_2$ 를 검정하기 위한 F-통계량을 구하여라.

$H_0 : \beta_1 = \beta_2 \Rightarrow A\beta = 0$, $A = [1 \; -1]$

$\dfrac{(SSR_R - SSR_U)/q}{SSR_U/(n-p-1)}$ * SSR : Residual sum of squares
U : unrestricted , R : restricted

For linear restriction,

$$F = \frac{(A\hat{\beta})^T[A(X^TX)^{-1}A^T]^{-1}A\hat{\beta}}{\hat{\sigma}^2}, \quad \hat{\sigma} = \frac{RSS}{n-p-1}, \quad A = [1 \; -1]$$

$$X^TX = \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 2 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 2 & -1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 5 \end{bmatrix}$$

$$(X^TX)^{-1} = \frac{1}{30}\begin{bmatrix} 5 & 0 \\ 0 & 6 \end{bmatrix} = \begin{bmatrix} 1/6 & 0 \\ 0 & 1/5 \end{bmatrix}, \quad X^Ty = \begin{bmatrix} y_1 + 2y_2 + y_3 \\ -y_2 + 2y_3 \end{bmatrix}$$

$$\hat{\beta} = (X^TX)^{-1}X^Ty = \begin{bmatrix} 1/6 & 0 \\ 0 & 1/5 \end{bmatrix}\begin{bmatrix} y_1 + 2y_2 + y_3 \\ -y_2 + 2y_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{6}(y_1 + 2y_2 + y_3) \\ -\frac{1}{5}(y_2 - 2y_3) \end{bmatrix}$$

$$A\hat{\beta} = [1 \; -1] \cdot \begin{bmatrix} \frac{1}{6}(y_1 + 2y_2 + y_3) \\ -\frac{1}{5}(y_2 - 2y_3) \end{bmatrix} = \frac{1}{6}y_1 + \left(\frac{1}{3} + \frac{1}{5}\right)y_2 + \left(\frac{1}{6} - \frac{2}{5}\right)y_3 = \frac{5y_1 + 16y_2 - 7y_3}{30}$$

$$A(X^TX)^{-1}A^T = [1 \; -1]\begin{bmatrix} 1/6 & 0 \\ 0 & 1/5 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix} = 1/6 + 1/5 = \frac{11}{30}$$

$$(A\hat{\beta})^T(A(X^TX)^{-1}A^T)^{-1}(A\hat{\beta}) = \frac{(5y_1+16y_2-7y_3)^2}{30\cdot 11}$$

$$RSS = y^T(I_n-\Pi_X)y = y^Ty - y^TX(X^TX)^{-1}X^Ty$$

$$= y^Ty - y^TX\hat{\beta}$$

$$\begin{array}{cc} & 1\times 2 \quad\quad 2\times 1 \end{array}$$

$$= y_1^2+y_2^2+y_3^2 - [y_1\ y_2\ y_3]\cdot\begin{bmatrix}1&0\\2&-1\\1&2\end{bmatrix}\begin{bmatrix}\frac{1}{6}(y_1+2y_2+y_3)\\ -\frac{1}{5}(y_2-2y_3)\end{bmatrix}$$

$$= y_1^2+y_2^2+y_3^2 - \left\{\frac{1}{6}(y_1+2y_2+y_3)^2 + \frac{1}{5}(y_2-2y_3)^2\right\}$$

$$= y_1^2+y_2^2+y_3^2 - \left\{\frac{1}{6}(y_1^2+4y_2^2+y_3^2+6y_1y_2+3y_1y_3+6y_2y_3) + \frac{1}{5}(y_2^2-4y_2y_3+4y_3^2)\right\}$$

$$F = \frac{(A\hat{\beta})^T(A(X^TX)^{-1}A^T)^{-1}(A\hat{\beta})/q}{SSR/(n-K)} = \frac{(5y_1+16y_2-7y_3)^2}{SSR\cdot 30\cdot 11\cdot 2} \sim F(2,1)$$

Q 7.10) 선형회귀모형

$$y_j = \beta_0+\beta_1 x_{1j}+\cdots+\beta_K x_{Kj}+\varepsilon_j \quad , j=1,2,\cdots,n$$

에서 가설 $H_0: \beta_i = c$를 검정하기 위한 F통계량을 구하여라.

$$\beta_i = c \Rightarrow \beta = \begin{bmatrix}\beta_1\\ \vdots\\ \beta_K\end{bmatrix} = \begin{bmatrix}\beta_1\\ \vdots\\ c\\ \vdots\\ \beta_K\end{bmatrix}$$

Let $\beta_c = [\beta_1 \cdots c \cdots \beta_K]^T$

Full-Model: $y_j = \beta_0+\beta_1x_{1j}+\cdots+\beta_Kx_{Kj}+\varepsilon_j$

Sub-Model: $y_j = \beta_0+\beta_1x_{1j}+\cdots+cx_{ij}+\cdots+\beta_Kx_{Kj}+\varepsilon_j$

$y_j - cx_{ij} = \beta_0+\beta_1x_{1j}+\cdots+\beta_Kx_{Kj}+\varepsilon_j$

Full-model: $y_j - cx_{ij} = \beta_0+\beta_1x_{1j}+\cdots+(\beta_i-c)x_{ij}+\cdots+\beta_Kx_{Kj}+\varepsilon_j$

Sub-model: $y_j - cx_{ij} = \beta_0+\beta_1x_{1j}+\cdots+\beta_{i-1}x_{i-1j}+\beta_{i+1}x_{i+1j}+\cdots+\beta_Kx_{Kj}+\varepsilon_j$

Let $\delta = \beta_i - c$. Then,

Full-model: $y_j - cx_{ij} = \beta_0+\beta_1x_{1j}+\cdots+\delta x_{ij}+\cdots+\beta_Kx_{Kj}+\varepsilon_j$

Sub-model: $y_j - cx_{ij} = \beta_0+\beta_1x_{1j}+\cdots+\beta_{i-1}x_{i-1j}+\beta_{i+1}x_{i+1j}+\cdots+\beta_Kx_{Kj}+\varepsilon_j$

$\Rightarrow$ Full-model: $y - cX_2 = \gamma_1X_1+\gamma_2X_2$

Sub-model: $y - cX_2 = \gamma_1X_1$

$\Rightarrow$ Full-model: $y^* = \gamma_1X_1+\gamma_2X_2$

Sub-model: $y^* = \gamma_1X_1$

$\gamma_1 = [\beta_0 \cdots \beta_{i-1}\ \beta_{i+1} \cdots \beta_K]^T$

$\gamma_2 = \beta_i$

$X_1 = [1\ \ x_1\ \cdots\ x_{i-1}\ \ x_{i+1}\cdots x_n]$

$X_2 = [x_i]$

$R(\gamma_2|\gamma_1) = y^{*T}\Pi_{X_{2,1}}y^*$

$SSE(\gamma_2|\gamma_1) = y^{*T}(I_n-\Pi_X)y^*$

$$\frac{R(\gamma_2|\gamma_1)}{SSE/(n-p-1)} = \frac{(y-cX_2)^T\Pi_{X_{2,1}}(y-cX_2)}{(y-cX_2)^T(I_n-\Pi_X)(y-cX_2)} \sim F(1,n-2)$$

$A\beta = t$ 꼴로서 $A = e_i^T$, $t = c$

$\hat{\beta} = (X^TX)^{-1}X^Ty$

$SSE = y^T(I_n-\Pi_X)y$

$$F = \frac{(e_i^T\hat{\beta}-c)^T(e_i^T(X^TX)^{-1}e_i)^{-1}(e_i^T\hat{\beta}-c)}{SSE/(n-p-1)}$$

7.11) $y = \theta+\epsilon$ 이고 $y' = (y_1, y_2, y_3, y_4)$, $\theta' = (\theta_1, \theta_2, \theta_3, \theta_4)$

$\epsilon' = (\epsilon_1, \epsilon_2, \epsilon_3, \epsilon_4)$ 이며 $\epsilon \sim N(0, I\sigma^2)$ 이다. 만약, $\theta_1+\theta_2+\theta_3+\theta_4 = 0$ 인 경우에

가설 $H_0: \theta_1 = \theta_3$을 검정하기 위한 F-통계량이

$$F_0 = \frac{2(y_1-y_3)^2}{(y_1+y_2+y_3+y_4)^2}$$

임을 증명하라.

$$\hat{\beta}_r = \underset{\substack{\beta\in\mathbb{R}^{p+1}\\ A\beta_r=0}}{\operatorname{argmin}} \|y - X\beta\|^2$$

$A\beta = 0 \Leftrightarrow A(X^TX)^{-1}X^TX\beta = 0$

Let $X_r^T = A(X^TX)^{-1}X^T$, $X_r = X(X^TX)^{-1}A^T$

$X\hat{\beta}_r = \Pi_Xy - \Pi_{X_r}y$

$= X(X^TX)^{-1}X^Ty - X_r(X_r^TX_r)^{-1}X_r^Ty$

$\therefore \hat{\beta}_r = (X^TX)^{-1}X^Ty - (X^TX)^{-1}A^T(A(X^TX)^{-1}A^T)^{-1}A(X^TX)^{-1}X^Ty$

$= \hat{\beta} - (X^TX)^{-1}A^T(A(X^TX)^{-1}A^T)^{-1}A\hat{\beta}$

$\theta_4 = -(\theta_1+\theta_2+\theta_3)$ 이므로 선형회귀 모형을 다음과 같이 쓸 수 있다.

$$\begin{bmatrix}y_1\\y_2\\y_3\\y_4\end{bmatrix} = \begin{bmatrix}1&0&0\\0&1&0\\0&0&1\\-1&-1&-1\end{bmatrix}\begin{bmatrix}\theta_1\\\theta_2\\\theta_3\end{bmatrix} + \begin{bmatrix}\epsilon_1\\\epsilon_2\\\epsilon_3\\\epsilon_4\end{bmatrix}$$

따라서

$$X'X = \begin{bmatrix}2&1&1\\1&2&1\\1&1&2\end{bmatrix}, \quad X^Ty = \begin{bmatrix}y_1-y_4\\y_2-y_4\\y_3-y_4\end{bmatrix}$$

이며, θ의 추정량은

$$\hat{\theta} = \begin{bmatrix}\hat{\theta}_1\\\hat{\theta}_2\\\hat{\theta}_3\end{bmatrix} = (X^TX)^{-1}X^Ty = \begin{bmatrix}y_1-\bar{y}\\y_2-\bar{y}\\y_3-\bar{y}\end{bmatrix}$$

이며,

$$SSE(F) = y'y - \hat{\theta}'X'X\hat{\theta} = \frac{1}{4}(\Sigma y_i)^2$$

이다.

귀무가설은 $H_0: C\theta = 0$ 이라고 표현할 때, $C = (1, 0, -1)$ 이므로,

$$Q = SSE(R) - SSE(F) = (C\hat{\theta})'[C(X^TX)^{-1}C']^{-1}C\hat{\theta} = \frac{1}{2}(y_1-y_3)^2$$

이 되므로, 검정통계량은

$$F_0 = \frac{Q/1}{SSE(F)/1} = \frac{\frac{1}{2}(y_1-y_3)^2}{\frac{1}{4}(\Sigma y_i)^2} = \frac{2(y_1-y_3)^2}{(y_1+\cdots+y_4)^2}$$

7.11) $y=\theta+\epsilon$ 이고 $y'=(y_1, y_2, y_3, y_4)$, $\theta'=(\theta_1, \theta_2, \theta_3, \theta_4)$

$\epsilon'=(\epsilon_1, \epsilon_2, \epsilon_3, \epsilon_4)$ 이며 $\epsilon \sim N(0, I\sigma^2)$ 이다. 만약, $\theta_1+\theta_2+\theta_3+\theta_4=0$ 인 경우에

가설 $H_0: \theta_1=\theta_3$ 을 검정하기 위한 F-통계량이

$$F_0=\frac{2(y_1-y_3)^2}{(y_1+y_2+y_3+y_4)^2}$$

임을 증명하라.

$$y=\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -1 & -1 \end{bmatrix}\begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix}+\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \epsilon_4 \end{bmatrix}$$

$$= X\beta+\epsilon$$

$A\beta=0$, $A=[1 \ 0 \ -1]$

$$X^TX=\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}+\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}=I_3+\mathbb{1}\mathbb{1}^T$$

$$(X^TX)^{-1}=(I_3+\mathbb{1}\mathbb{1}^T)^{-1}=I_3-\frac{\mathbb{1}\mathbb{1}^T}{1+\mathbb{1}^T\mathbb{1}}=I_3-\frac{1}{4}\mathbb{1}\mathbb{1}^T$$

$$=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}-\begin{bmatrix} 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 \end{bmatrix}=\begin{bmatrix} 3/4 & -1/4 & -1/4 \\ -1/4 & 3/4 & -1/4 \\ -1/4 & -1/4 & 3/4 \end{bmatrix}$$

$$X^TY=\begin{bmatrix} y_1-y_4 \\ y_2-y_4 \\ y_3-y_4 \end{bmatrix}$$

$$\therefore \hat{\theta}=(X^TX)^{-1}X^TY=\frac{1}{4}\begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}\begin{bmatrix} y_1-y_4 \\ y_2-y_4 \\ y_3-y_4 \end{bmatrix}=\frac{1}{4}\begin{bmatrix} 3y_1-3y_4-y_2+y_4-y_3+y_4 \\ -y_1+y_4+3y_2-3y_4-y_3+y_4 \\ -y_1+y_4-y_2+y_4+3y_3-3y_4 \end{bmatrix}$$

$$=\begin{bmatrix} y_1-\bar{y} \\ y_2-\bar{y} \\ y_3-\bar{y} \end{bmatrix}$$

$$F=\frac{(A\hat{\theta})^T(A(X^TX)^{-1}A^T)^{-1}(A\hat{\theta})}{SSE/(n-p-1)}$$

$$A(X^TX)^{-1}A^T=\frac{1}{4}[1 \ 0 \ -1]\begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$=\frac{1}{4}[4 \ 0 \ -4]\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}=\frac{1}{4}(4+4)=2$$

$\therefore (A(X^TX)^{-1}A^T)^{-1}=\frac{1}{2}$

$A\hat{\theta}=y_1-y_3 \quad \therefore (A\hat{\theta})^T(A(X^TX)^{-1}A^T)^{-1}(A\hat{\theta})=\frac{1}{2}(y_1-y_3)^2$

$SSE=y^T(I-\Pi_X)y$

$SSE=y^T(I_n-\Pi_X)y$

$$=y^Ty-\hat{y}^T\hat{y}=\Sigma y_i^2-\hat{\theta}^TX^TX\hat{\theta}$$

$$\hat{\theta}^T(X^TX)\hat{\theta}=\begin{bmatrix} y_1-\bar{y} \\ y_2-\bar{y} \\ y_3-\bar{y} \end{bmatrix}^T(I+\mathbb{1}\mathbb{1}^T)\begin{bmatrix} y_1-\bar{y} \\ y_2-\bar{y} \\ y_3-\bar{y} \end{bmatrix}$$

$$=\sum_1^3(y_i-\bar{y})^2+(y_1+y_2+y_3-3\bar{y})^2$$

$$=\sum_1^3(y_i-\bar{y})^2+(-y_4+\Sigma y_i-4\bar{y}+\bar{y})^2$$

$$=\sum_1^3(y_i-\bar{y})^2+(y_4-\bar{y})^2$$

$$=\sum_1^4(y_i-\bar{y})^2$$

$$=\sum_1^4(y_i^2-2y_i\bar{y}+\bar{y}^2)$$

$$=\Sigma y_i^2-8\bar{y}^2+4\bar{y}^2$$

$$\therefore y'y-\hat{\theta}^T\hat{\beta}^T\hat{\beta}\hat{\theta}=4\bar{y}^2=\frac{1}{4}(\Sigma y_i)^2$$

$$\therefore F=\frac{(A\hat{\beta})^T(A(X^TX)^{-1}A^T)^{-1}(A\hat{\beta})}{SSE/(n-p-1)}=\frac{\frac{1}{2}(y_3-y_1)^2}{\frac{1}{4}(\Sigma y_i)^2/(4-3)}$$

$$=\frac{2(y_3-y_1)^2}{(\Sigma y_i)^2}$$